

Assignment no : 05

Roll no : 13216

Problem Statement : Data Analytics II

1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Load Dataset

```
dataset = pd.read_csv("Social_Network_Ads (1).csv")
print(dataset.head())
```

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

Select Features (X) and Target (y)

```
X = dataset.iloc[:, [2, 3]].values    # Age & Salary
y = dataset.iloc[:, 4].values         # Purchased (0/1)
```

Split Dataset (Training & Testing)

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=0)
```

Feature Scaling (Important)

```
from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
```

Apply Logistic Regression

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression()
model.fit(X_train, y_train)

LogisticRegression()
```

Predict Results

```
y_pred = model.predict(X_test)
```

Create Confusion Matrix

```
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)
print(cm)

[[65  3]
 [ 8 24]]
```

Extract Values

```
TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

accuracy = (TP + TN) / (TP + TN + FP + FN)
error_rate = 1 - accuracy
precision = TP / (TP + FP)
recall = TP / (TP + FN)

print("TP:", TP)
print("TN:", TN)
print("FP:", FP)
print("FN:", FN)

print("Accuracy:", accuracy)
print("Error Rate:", error_rate)
print("Precision:", precision)
print("Recall:", recall)

TP: 24
TN: 65
FP: 3
```

FN: 8
Accuracy: 0.89
Error Rate: 0.10999999999999999
Precision: 0.8888888888888888
Recall: 0.75

```
from sklearn.metrics import classification_report
```

```
print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	0.89	0.96	0.92	68
1	0.89	0.75	0.81	32
accuracy			0.89	100
macro avg	0.89	0.85	0.87	100
weighted avg	0.89	0.89	0.89	100

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay  
import matplotlib.pyplot as plt
```

```
# Create confusion matrix
```

```
cm = confusion_matrix(y_test, y_pred)
```

```
# Display diagram
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm,  
display_labels=["Not Purchased", "Purchased"])  
disp.plot(cmap='Blues')
```

```
plt.title("Confusion Matrix Diagram")
```

```
plt.show()
```

